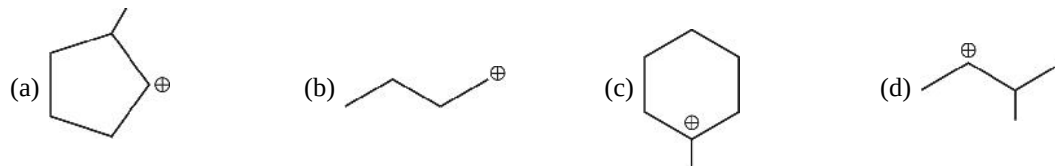


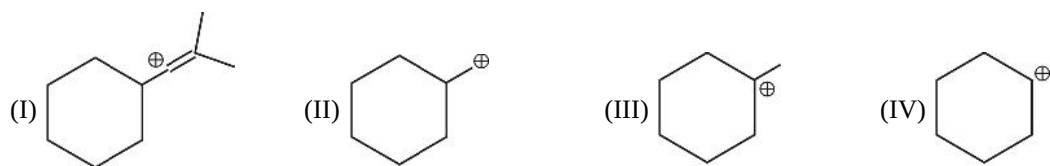
LDA - 12

Carbocation

1. Which of the following would not rearrange to a more stable form ?



2. List the following carbocations in order of decreasing stabilization energies



- (a) II, III, I, IV (b) III, IV, II, I (c) III, IV, I, II (d) I, II, IV, III

3. (I) CH_3^+ (II) $\text{CH}_3\text{—CH}_2^+$ (III) $\text{CH}_3\text{—CH}^+\text{—CH}_3$ (IV) $\text{CH}_3\text{—C}^+(\text{CH}_3)_2$
 (1°) (2°) (3°)

The stability order among these alkyl cations is

- (a) I > II > III > IV (b) IV > III > II > I (c) III > IV > II > I (d) II > III > IV > I

4. (I) $\text{CH}_3\text{—CH}^+\text{—CH}_2\text{—CH}_2\text{—CH}_3$ (II) $\text{CH}_3\text{—CH}_2\text{—CH}^+\text{—CH}_2\text{—CH}_3$
 (III) $\text{CH}_3\text{—CH}_2\text{—C}^+(\text{CH}_3)\text{—CH}_2\text{—CH}_3$

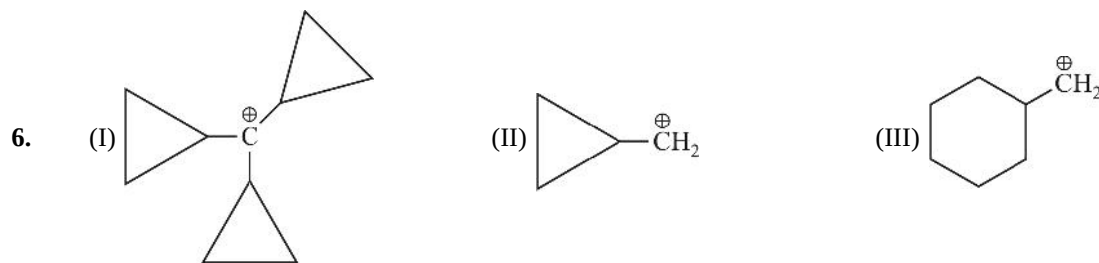
Which of the following orders is correct for the stability of these carbocations ?

- (a) I > II > III (b) III > II > I (c) III > I > II (d) I > III > II

5. (I) $\text{CH}_3\text{—CH}_2^+$ (II) $\text{CH}_2=\text{CH}^+$ (III) $\text{HC}\equiv\text{C}^+$

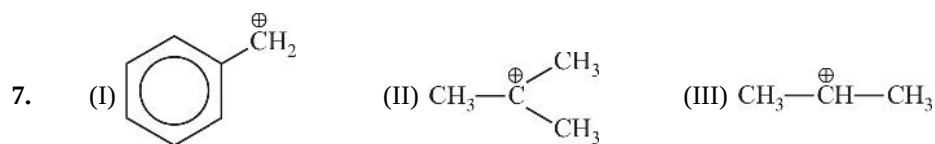
Which of the following orders is correct for the stability of these carbocations ?

- (a) I > II > III (b) III > II > I (c) I > III > II (d) II > III > I



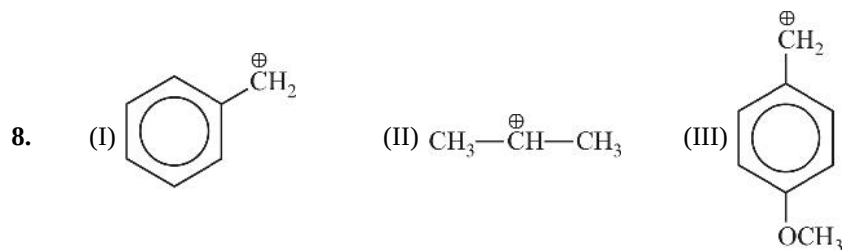
Which of the following orders is correct for the stability of these carbocations ?

- (a) I > II > III (b) III > II > I (c) II > III > I (d) III > I > II



Which of the following orders is correct for the stability of these carbocations ?

- (a) I > II > III (b) III > II > I (c) II > I > III (d) II > I $\approx$ III



Which of the following orders is correct for the stability of these carbocations ?

- (a) I > II > III (b) III > II > I (c) III > I > II (d) III $\approx$ I $\approx$ II

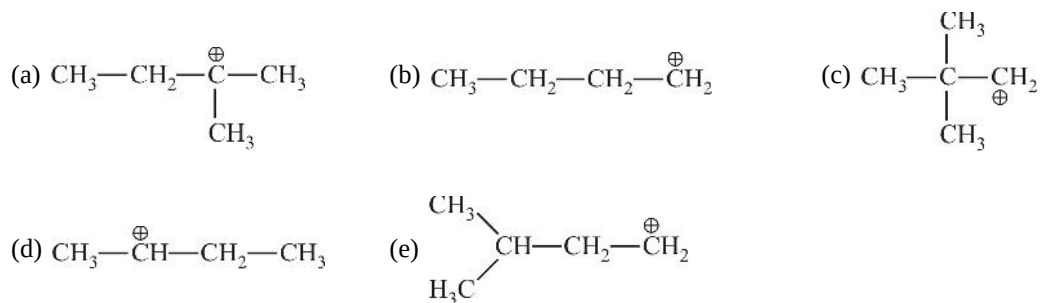
9. Which of the following statements is correct about carbocations ?

- (a) A less stable carbocation can rearrange itself into a more stable carbocation, if structure permits.
 (b) In rearrangement, an atom or group migrates from the adjacent carbon to positive centre.
 (c) This rearrangement, can lead to contraction of larger rings and expansion of smaller rings.
 (d) All of these

10. What is the effect of introduction of an alkyl group on the stability of the carbocation ?

11. Why allyl cation is more stable than ethyl cation ?

12. Identify each of following carbocations as being either 1°, 2° or 3°.



13. Which of the two reactions shown below would be faster ? Why ?



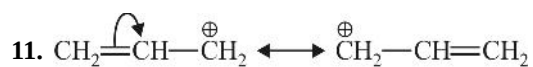
14. Which would you expect to be more stable – $(\text{CH}_3)_3\text{C}^+$ or $(\text{CF}_3)_3\text{C}^+$? Why ?

15. Rank the following carbocation in decreasing order of stability



Answer Key

1. (c) 2. (b) 3. (b) 4. (c) 5. (a) 6. (a) 7. (a) 8. (c)
9. (d) 10. +I group (alkyl group) increases the stability of carbocation.



Allyl carbocation is resonance stabilized.

12. (a) 3° (b) 1° (c) 1° (d) 2° (e) 1°

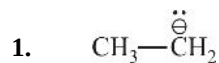
13. 'A' leads to formation of more stable carbocation.

14. $(\text{CH}_3)_3\text{C}^{\oplus}$; +I effect of CH_3 .

15. II > I

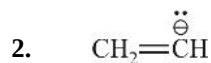
LDA - 13

Carbanion & Carbon free radical



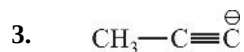
Lone-pair of electrons of this carbanion is in which of the following orbitals ?

- (a) 2p (b)
- sp^3
- (c)
- sp^2
- (d) sp



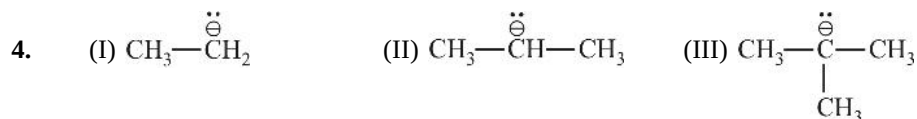
Lone pair of electrons of this carbanion is in which of the following orbitals ?

- (a) 2p (b)
- sp^3
- (c)
- sp^2
- (d) sp



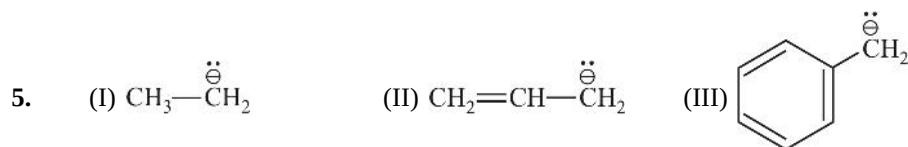
Lone-pair of electrons of this carbanion is in which of the following orbitals ?

- (a) 2p (b)
- sp^3
- (c)
- sp^2
- (d) sp



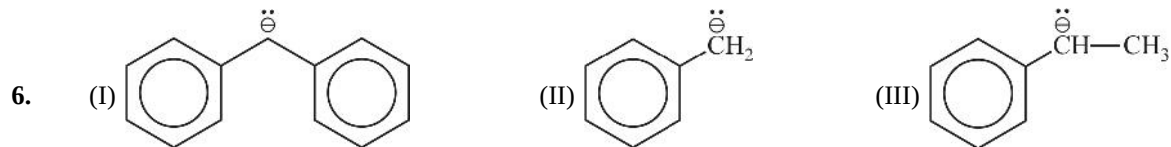
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I (c) II > I > III (d) I
- $\approx$
- II
- $\approx$
- III



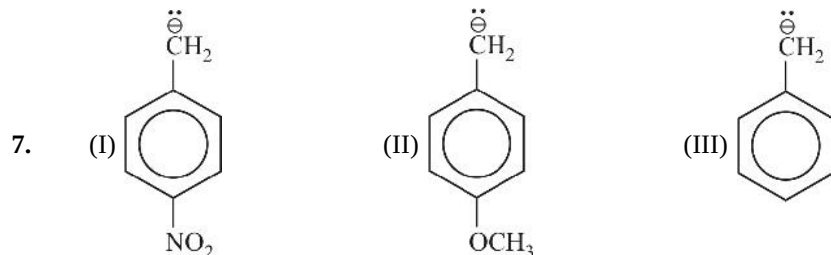
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I (c) II > III > I (d) II > I > III



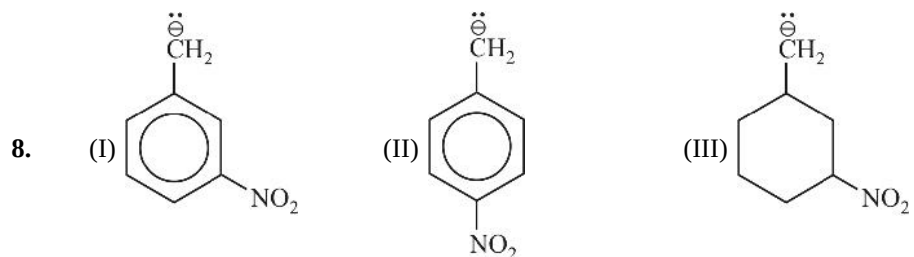
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I (c) II > III > I (d) III > I > II



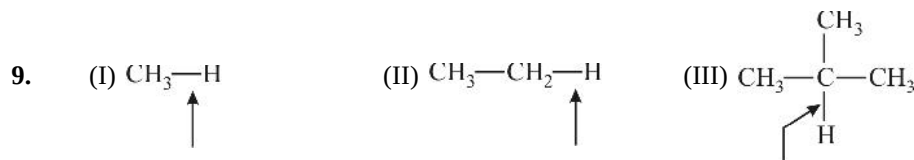
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I (c) I > III > II (d) II > III > I



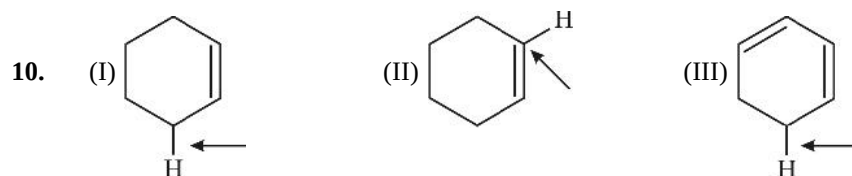
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I (c) II > I > III (d) II > III > I



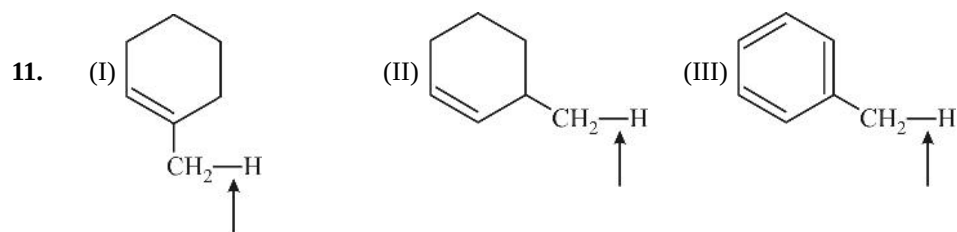
Which of the following orders is correct for the energy required for homolytic cleavage of indicated C-H bonds ?

- (a) I > II > III (b) III > II > I (c) III > I > II (d) II > I > III



Which of the following orders is correct for the energy required for homolytic cleavage of indicated C-H bonds ?

- (a) I > II > III (b) III > II > I (c) II > I > III (d) II > III > I



Which of the following orders is correct for the energy required for homolytic cleavage of indicated C-H bonds ?

- (a) I > II > III (b) III > II > I (c) II > I > III (d) III > I > II



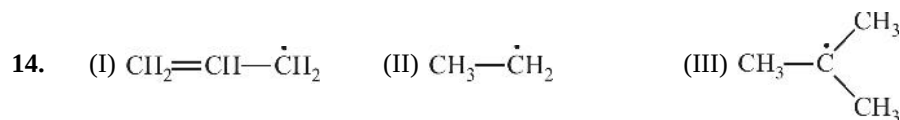
Hybridization of the C-atom of this radical is

- (a) sp^3 (b) sp^2 (c) sp (d) unhybridized



Hybridization of the C-atom of this radical is

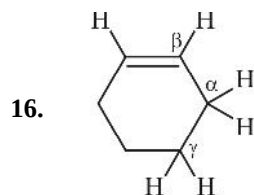
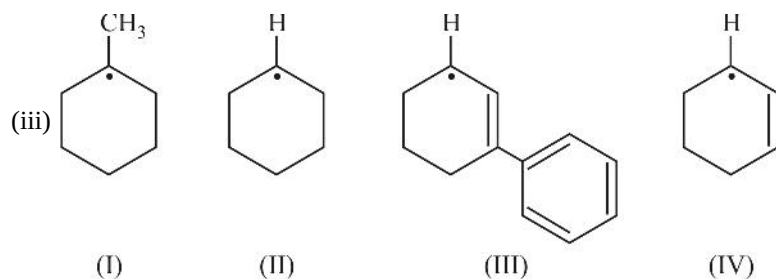
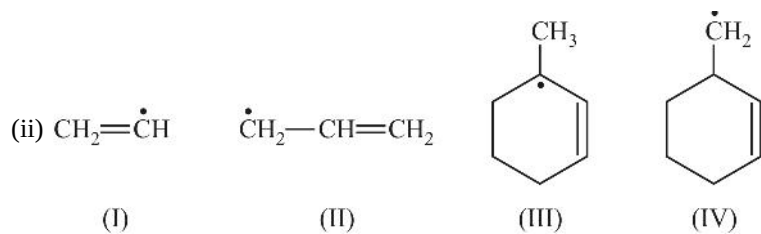
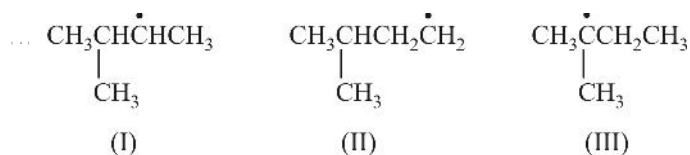
- (a) sp^3 (b) sp^2 (c) sp (d) unhybridized



Which of the following orders is correct for the stability of these radicals ?

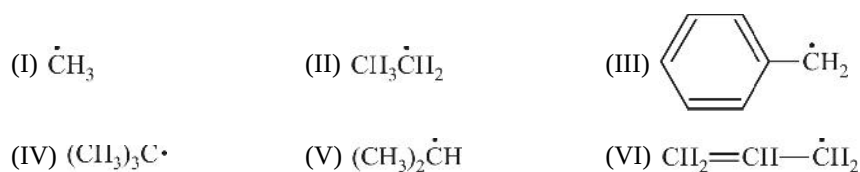
- (a) I > II > III (b) III > II > I (c) I > III > II (d) III > I > II

15. Arrange the following in increasing order of stability



Arrange the BE of (C-H) bonds indicated by α , β and γ .

17. Rank the following radicals in decreasing order of stability



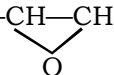
Answer Key

1. (b) 2. (c) 3. (d) 4. (a) 5. (b) 6. (a) 7. (c) 8. (c) 9. (a)
 10. (c) 11. (c) 12. (b) 13. (a) 14. (c)
 15. (i) A : II < I < III (ii) C : I < IV < II < III (iii) D : II < I < IV < III 16. $\beta > \gamma > \alpha$
 17. III > VI > IV > V > II > I

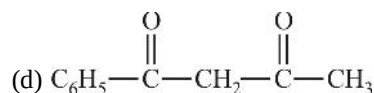
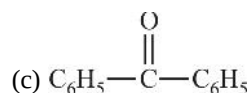
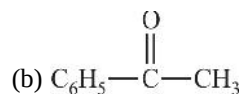
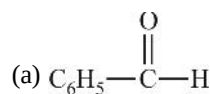
LDA - 14

Structural Isomerism

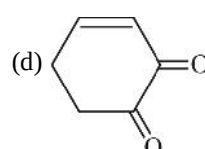
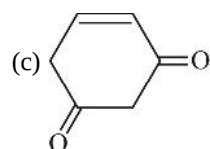
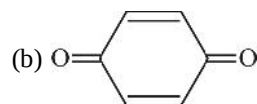
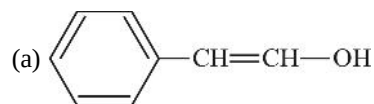
- The isomerism observed in alkanes is :
 (a) metamerism (b) chain isomerism (c) position isomerism (d) geometrical isomerism
- Which of the following are functional isomers ?
 (a) $\text{CH}_3\text{CH}_2\text{Cl}$ and $\text{CH}_3\text{CH}_2\text{Br}$ (b) CH_3CHBr_2 and $\text{CH}_2\text{Br} \cdot \text{CH}_2\text{Br}$
 (c) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ and $\text{CH}_3\text{OC}_3\text{H}_7$ (d) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3-\text{CH}-\text{CH}_2$



- Keto-enol tautomerism is observed in

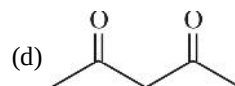
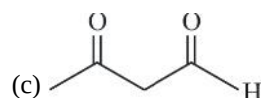
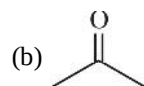


- Tautomerism is exhibited by

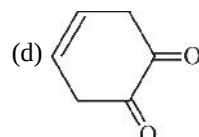
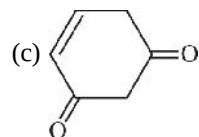
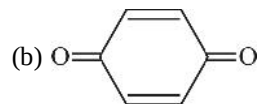
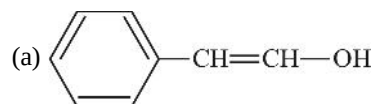


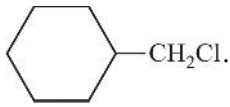
- The enolic form of acetone contains
 (a) 9 σ bonds, 1 π bond and 2 lone pairs
 (b) 8 σ bonds, 2 π bonds and 2 lone pairs
 (c) 10 σ bonds, 1 π bond and 1 lone pair
 (d) 9 σ bonds, 2 π bonds and 1 lone pair
- Which of the following has maximum enol content ?

- (a) Acetaldehyde



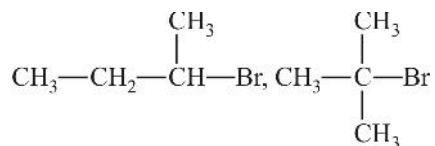
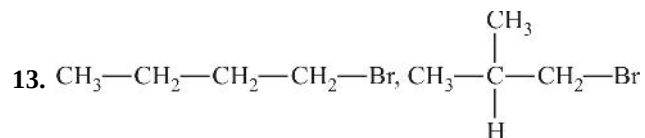
- Tautomerism is not exhibited by



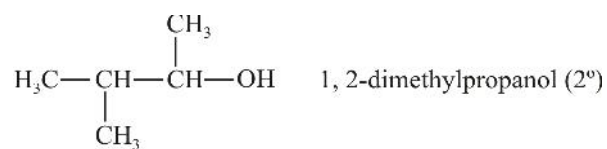
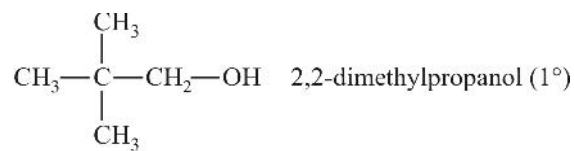
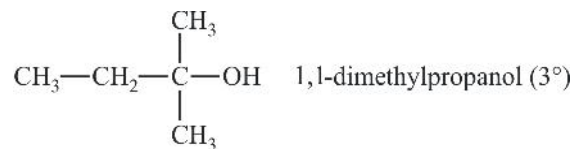
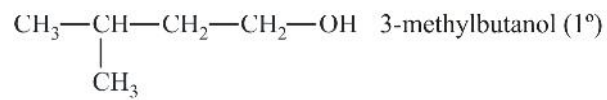
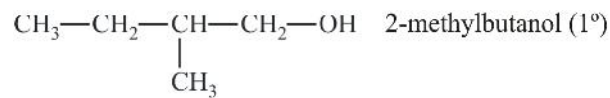
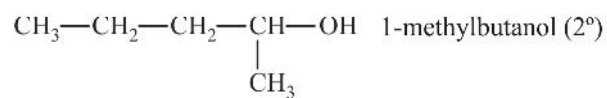
8. Which organic structure among the following is not an isomer of the compound ?
 $\text{CH}_3\text{—CO—CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$?
 (a) $\text{CH}_3\text{CH=CHCH}_2\text{CH}_2\text{CHO}$ (b) $\text{CH}_3\text{CH}_2\text{OCH=CHCH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_2\text{CH}_3$ (d) $(\text{CH}_3)_2\text{CH—CO—CH}_2\text{CH}_3$
9. The compound which is not isomeric with diethyl ether is
 (a) n – propylmethyl ether (b) butanol – 1
 (c) 2 – methylpropanol (d) butanone
10. The maximum number of an alkene with molecular formula C_4H_8 is
 (a) 2 (b) 3 (c) 4 (d) 5
11. How many cyclic isomer of C_5H_{10} are possible ?
 (a) 4 (b) 3 (c) 2 (d) 5
12. How many chain isomers can be obtained from the alkane C_6H_{14} ?
 (a) 3 (b) 5 (c) 6 (d) 7
13. Write the structural isomers of the compound having formula $\text{C}_4\text{H}_9\text{Br}$.
14. (i) Draw the structures of all isomeric alcohols of molecular formula $\text{C}_5\text{H}_{12}\text{O}$ and give their IUPAC names.
 (ii) Classify the isomers of alcohols in question as primary, secondary and tertiary alcohols.
15. Write position isomers of .

Answer Key

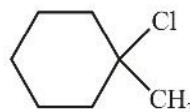
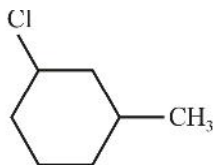
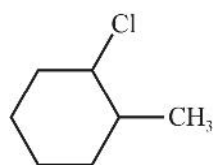
1. (a,b,c) 2. (d) 3. (b,d) 4. (a,c,d) 5. (a) 6. (c) 7. (b) 8. (a) 9. (d)
 10. (b) 11. (a) 12. (a)



14. (i) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$ pentanol (1°)



15.



LDA - 15

Geometrical Isomerism & Conformers

1. Which of the following compounds will show cis-trans isomerism ?

- (i) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{C}_2\text{H}_5$ (ii) $\text{CH}_2=\text{CBr}_2$
 (iii) $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{CH}_3$ (iv) $\text{CH}_3\text{CH}=\text{CClCH}_3$

2. Which of the following compounds will exhibit cis-trans (geometrical) isomerism ?

- (a) 2-Butene (b) 2-Butyne (c) 2-Butanol (d) Butanol

3. Geometrical isomerism is shown by



4. Which of the following compounds will show geometrical isomerism ?

- (a) 2-Butene (b) Propene (c) 1-Phenylpropene (d) 2-Methyl-2-butene

5. Which of the following compounds will exhibit geometrical isomerism ?

- (a) 1-Phenyl-2-butene (b) 3-Phenyl-1-butene (c) 2-Phenyl-1-butene (d) 1, 1-Diphenyl-1-propene

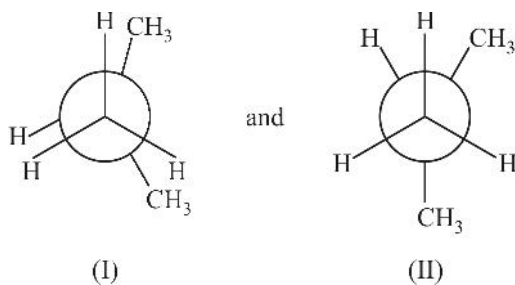
6. Which of the following compounds will show geometrical isomerism ?

- (a) 1-Butane (b) Propane (c) 1-Phenyl propene (d) 2-Methyl-2-butene

7. Which of the following isomerism is not found in alkenes ?

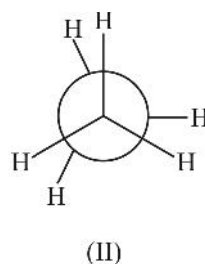
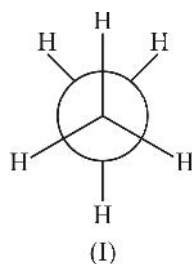
- (a) Metamerism (b) Geometrical isomerism (c) Position isomerism (d) Chain isomerism

8. What are the type of isomers in following pairs ?



- (a) conformers (b) position (c) optical (d) geometrical

9. Which type of conformation is shown by I and II ?



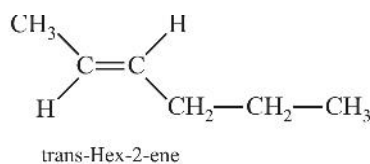
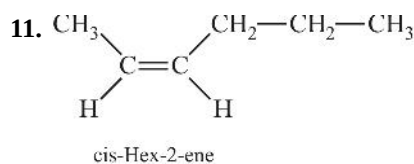
- (a) I is eclipsed, II is staggered
(b) II is eclipsed, I is staggered
(c) both are eclipsed
(d) both are staggered
10. Number of which of the following isomers is always infinite ?
(a) Geometrical isomers
(b) Optical isomers
(c) Conformational isomers
(d) Structural isomers
11. Draw the cis and trans structures of hex-2-ene. Which isomer will have higher b.p. and why ?
12. Draw cis and trans isomers of the following compounds. Also write their IUPAC names
(i) $\text{CHCl}=\text{CHCl}$
(ii) $\text{C}_2\text{H}_5\text{CCH}_3=\text{CCH}_3\text{C}_2\text{H}_5$
13. 2,4-hexadiene has three geometrical isomers. Draw their structures.
14. What are eclipsed and staggered conformations of ethane and explain their relative stability ?
15. What do you understand by the term conformations ? How many conformations of ethane are possible in all ?

Answer Key

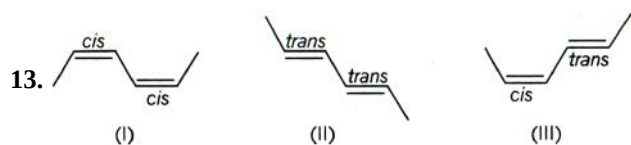
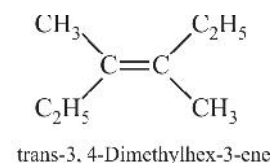
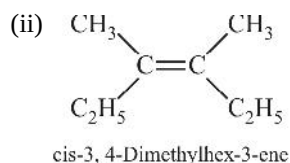
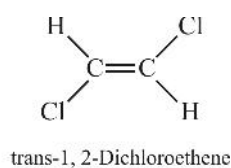
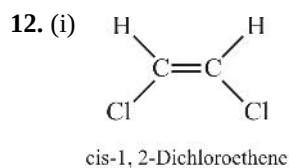
1. (iii) and (iv). In structures (i) and (ii), two identical groups are attached to one of the doubly bonded carbon atom.

2. (a) 3. (b) 4. (ac) 5. (a) 6. (c) 7. (a) 8. (a) 9. (b)

10. (c)



The cis form will have higher boiling point due to more polar nature leading to stronger intermolecular dipole-dipole interaction, thus requiring more heat energy to separate them.



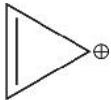
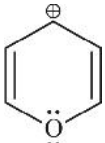
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Aromaticity

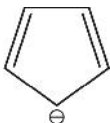
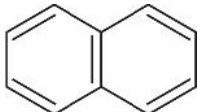
1. Which of the following is not the Huckel number ?

- (a) 6 (b) 2 (c) 8 (d) 14

2. Which of the following species should be aromatic by the Huckel rule ?

- (a)  (b)  (c)  (d) all of these

3. Which one of the following substances is not aromatic ?

- (a)  (b)  (c)  (d) 

4. 
(I) (II) (III)

Which of these cyclopropene systems is aromatic ?

- (a) I (b) II (c) III (d) all of these

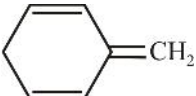
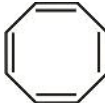
5. 
(I) (II) (III)

Which of these cyclopropene systems is anti-aromatic ?

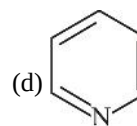
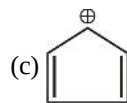
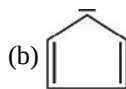
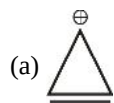
- (a) I (b) II (c) III (d) all of these

6. What are the necessary conditions for any system to be aromatic ?

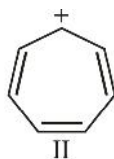
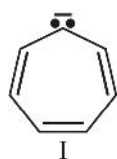
7. Explain why the following systems are not aromatic ?

- (i)  (ii)  (iii) 

8. Which of the following compounds is not aromatic ?



9. Which of the following species could be expected to exhibit aromatic character ?



Select the correct answer from the following.

(a) I and IV

(b) II and IV

(c) I and III

(d) II and III

10. An aromatic compound among other things should have a π -electron cloud containing $(4n + 2)$ π -electrons where n can't be

(a) $1/2$

(b) 3

(c) 2

(d) 1

11. Which one of the following is aromatic ?

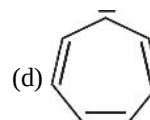
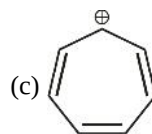
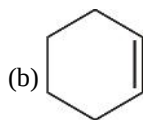
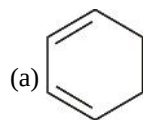
(a) Cyclopentadienyl cation

(b) Cyclo-octatetraene

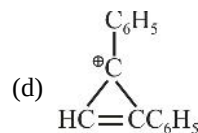
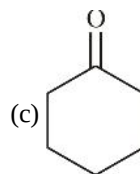
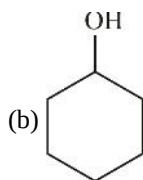
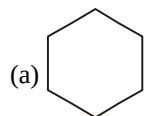
(c) Cycloheptatriene

(d) Cycloheptatrienyl cation

12. Which of the following is aromatic ?



13. Which of the following compound is aromatic ?



14. Nitrogen atom of pyrrole is

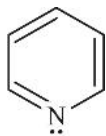
(a) sp^3 -hybridized

(b) sp^2 -hybridized

(c) sp -hybridized

(d) not hybridized

15.



Nitrogen atom of pyridine is

- (a) sp^3 hybridized (b) sp^2 hybridized (c) sp hybridized (d) not hybridized

Answer Key

1. (c)

2. (b)

3. (b)

4. (c)

5. (b)

8. (c)

9. (d)

10. (a)

11. (d)

12. (c)

13. (d)

14. (b)

15. (b)